Chapter 6
Bioinformatics in Immunoglobulin Glycosylation Analysis



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Contents

6.1 Introduction 206
6.2 Glycomic Data Collection and Processing 208
6.2.1 Reference Databases 208
6.2.2 Identification and Quantification Software Tools 210
6.3 Glycoproteomics Data Collection and Processing 210
6.3.1 Reference Databases 210
6.3.2 Identification Software Tools 211
6.3.3 Quantification Software Tools 212
6.4 Data Integration with Other Omics 212
6.5 Practical Examples 214
6.5.1 Glycomic Data Processing 214
6.5.2 Glycopeptide Data Processing for Enriched Immunoglobulins 218
6.5.3 Visualizing Profiles 224
6.6 Conclusion 228
References 231

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Abstract Analytical methods developed for studying immunoglobulin glycosylation rely heavily on software tailored for this purpose. Many of these tools are now used in high-throughput settings, especially for the glycomic characterization of IgG. A collection of these tools, and the databases they rely on, are presented in this chapter. Specific applications are detailed in examples of immunoglobulin glycomics and glycoproteomics data processing workflows. The results obtained in the glycoproteomics workflow are emphasized with the use of dedicated visualizing tools. These tools enable the user to highlight glycan properties and their differential expression.

Keywords Glycomics · Glycoproteomics · Workflow · Software · Database · Glycoinformatics

Abbreviations

ETD	Electron-transfer dissociation
Fc	Fragment crystallizable
HTP	High throughput
Ig	Immunoglobulin
IMGT	ImMunoGeneTics information system
LC	Liquid chromatography
MIRAGE	Minimum Information Required for A Glycomics Experiment
MS	Mass spectrometry
MS/MS	Tandem mass spectrometry
NMR	nuclear magnetic resonance
PDB	Protein Data Bank
PNGase F	Peptide-*N*-glycosidase F
RP	Reverse phase
RT	Retention time
SNFG	Symbol Nomenclature for Glycans

6.1 Introduction

Glycosylation plays a major role in creating proteoform diversity reaching the magnitude of 10^6 in the human proteome. The mapping of this diversity is far from being charted, particularly in the case of glycoforms, mainly because of the limitations and the very broad variety of experimental techniques needed to capture information. Mass spectrometry (MS) and nuclear magnetic resonance (NMR) as part of a fuller range of analytical methods (Gray et al. 2019) are used to solve the structure of glycans. In recent years, the improved accuracy of mass spectrometers and sophisticated fragmentation techniques continue to refine the identification of

intact glycopeptides (peptide with attached glycan) and new glycoproteomics datasets are regularly published (Ye et al. 2019; Riley et al. 2019).

Glycomics and glycoproteomics have benefitted from past experience in proteomics and the principles of the Proteome Standards Initiative (Orchard et al. 2003) and ProteomeXchange (Vizcaíno et al. 2014) are being gradually applied. In the past decade, reporting guidelines have been issued through the MIRAGE (Minimum Information Required for A Glycomics Experiment) initiative (https://www.beilstein-institut.de/en/projects/mirage/) as part of the many minimum information standards defined to facilitate data checking and processing (Kolarich et al. 2013).

Determining a glycosylation profile may entail either glycomics or glycoproteomics studies or both, depending on experimental design. In most cases, mass spectrometry data analysis software is used. Despite the expansion of glycan (Fujita et al. 2020) and glycoconjugate databases and repositories (Alocci et al. 2019; York et al. 2020), direct use of such resources is limited for glycan identification as there is no template that can be used to infer structures detectable in an organism. Some of the commonly applied strategies rely on established proteomics search tools that are refined to identify multiple modifications or exploiting idiosyncratic fragmentation patterns of glycans, while others use dedicated software for glyco(proteo)mics data, as exemplified in this chapter.

Glycosylation analyses of proteins can be performed either at intact glycoprotein, at glycopeptide, or released glycan level. Irrespective of the level of analytical method, structural analysis of glycoproteins remains elusive due to lack of high-throughput analytical tools and data analysis pipeline capable of analyzing high-volume spectral data to determine the glycan composition, structure and/or glycopeptide sequence information (Pralow et al. 2020; Alagesan et al. 2020). Immunoglobulins (Igs) and Fc receptors are complex glycoproteins and key components of both the innate and adaptive immune systems. The specific glycosylation of both immunoglobulin and its receptors is well known to be crucial for maintaining and modulating effector functions. In humans, there are five distinct antibody isotypes: IgM, IgD, IgE, IgA, and IgG. All of them are glycosylated (Arnold et al. 2007; de Haan et al. 2020). IgG provides an ideal case scenario for the development of high-throughput analytical tools and data analysis pipeline as it (1) is one of the most abundant proteins in human serum, accounting for about 10–20% of plasma protein, and (2) mostly bears a single *N*-linked glycan attached to Asn-297 of each heavy chain, which can have a wide variety of different structures. These unique IgG features allow for fine-tuning and extensive benchmarking of high-throughput (HTP) glycoanalytical methods (Trbojević-Akmačić et al. 2016; Wuhrer et al. 2007; Gstöttner et al. 2020; Amez-Martín et al. 2020).

This chapter briefly surveys the landscape of glycan and intact glycopeptide identification and quantification bioinformatics resources and focuses on profiling of antibodies using glycomics or glycoproteomics approaches as well as using visualization tools to easily compare these profiles. The coverage of resources is depicted in Fig. 6.1.

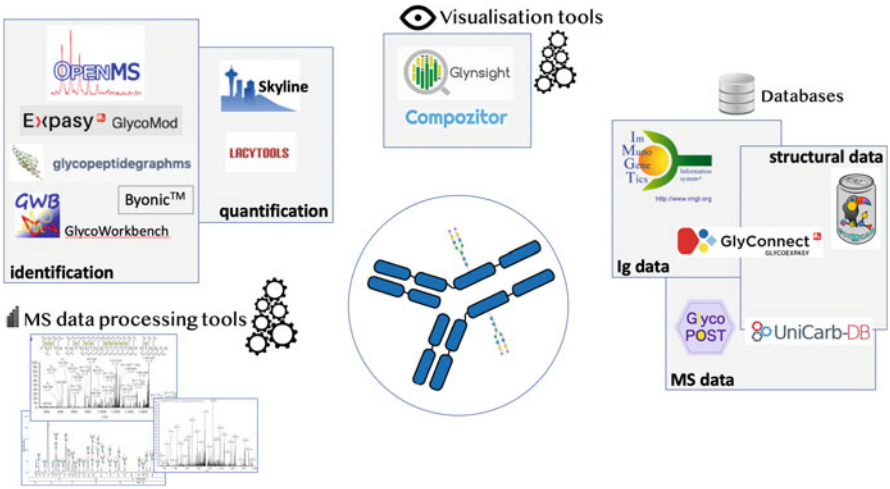


Fig. 6.1 Summary of resources. The glycoinformatics resources that are cited and used in this chapter are categorized as either database or tool. The tools are further distinguished according to their purpose: tools processing mass spectrometry (MS) data (identification or quantification) or used to visualize processed data. Databases store different types of data

6.2 Glycomic Data Collection and Processing

A recently published chapter covers widely the database and tool collection available for glycomics studies (Lisacek et al. 2017) presented as exhaustive lists of the various databases and software tools. The present section is an update on recent additions to the collection that is not reproduced here but complemented by the partially overlapping content of Table 6.1.

6.2.1 Reference Databases

In recent years, several community-driven moves in glycan bioinformatics have been initiated, the first of which is the agreement to resort to a single repository for glycan structures. This resulted in the creation of GlyTouCan in 2016, only briefly mentioned in (Lisacek et al. 2017) as version 1.0 (Aoki-Kinoshita et al. 2016). GlyTouCan is an uncurated registry for glycan structures that assigns globally unique accession numbers. It is increasingly adopted as a reference (Tiemeyer et al. 2017) and reached version 3.0 in 2021 to include a substantially reshaped registration flow improving data submission (Fujita et al. 2020). Manually annotated and experimentally verified spectra collected in UniCarb-DB (Hayes et al. 2011) are cross-referenced to GlyTouCan in order to increase the speed and accuracy of glycan assignment.

Table 6.1 Summary of cited glycoinformatics resources

Type	Resource name	Availability	References
Databases			
Glycoproteins	GlyConnect	glyconnect.expasy.org	Alocci et al. (2019)
Immunoglobulins	IMGT	imgt.org	Lefranc et al. (2015)
Glycans	UniCarb-DB	unicarb-db.expasy.org	Hayes et al. (2011)
3D structures	Protein Data Bank	www.wwpdb.org	wwPDB consortium et al. (2019)
Proteins	UniProt	uniprot.org	The UniProt Consortium (2019)
Repositories (Enabled data submission)			
MS data	GlycoPOST	glycopost.glycosmos.org	Watanabe et al. (2020)
Glycans	GlyTouCan	glytoucan.org	Fujita et al. (2020)
Software			
Generic MS Processing	ProteoWizard	proteowizard.sourceforge.net/tools.shtml	Kessner et al. (2008)
	OpenMS	www.openms.de	Pfeuffer et al. (2017)
Glycan MS Processing	GlycoMod	web.expasy.org/glycomod	Cooper et al. (2001)
	GlycoWorkbench	code.google.com/archive/p/glycoworkbench	Ceroni et al. (2008)
	GRITS toolbox	www.grits-toolbox.org	Weatherly et al. (2019)
Glycopeptide MS processing	Byonic	Licensed	Bern et al. (2012)
	GlycopeptideGraphMS	bitbucket.org/glycoaddict/glycopeptidegraphms/src/master/	Choo et al. (2019)
Quantitative MS processing	Skyline	skyline.ms/project/home/begin.view?	Pino et al. (2020)
	LacyTools	git.lumc.nl/cpm/lacytools	Jansen et al. (2016)
Visualization	Glynsight	glycoproteome.expasy.org/glynsight	Alocci et al. (2018)
	Compozitor	glyconnect.expasy.org/compozitor	Robin et al. (2020)

Glycomics mass spectrometry data have been taken one step further with the introduction of a MIRAGE-compliant experimental data repository named GlycoPOST (Watanabe et al. 2020) and the implementation of a pipeline for collecting such data (Rojas-Macias et al. 2019). Some publishers have started imposing data deposition and the field of glycoscience is slowly (re)connecting to life sciences.

6.2.2 Identification and Quantification Software Tools

GlycoMod (Cooper et al. 2001) and GlycoWorkbench (Ceroni et al. 2008) are still in use in much the same way for the identification of glycan structures, described in (Lisacek et al. 2017). These earlier protocols (1) from analytical MS data to monosaccharide composition using GlycoMod, (2) from structure to predicted MS/MS data using GlycoWorkbench, and (3) from analytical MS and MS/MS data to structure using UniCarb-DB, are still valid. A more recent identification platform is the GRITS toolbox (Weatherly et al. 2019) that can be considered as an upgrade of GlycoWorkBench. In particular, managing adducts is made more flexible and larger datasets composed of thousands of spectra are more easily handled.

Quantification was barely addressed in (Lisacek et al. 2017) mainly due to the limited number of quantitative glycomics datasets a few years back. The situation is rapidly changing and the need for related software is increasing. Skyline is accurately described by its authors as a software ecosystem for quantitative mass spectrometry informatics (Pino et al. 2020). This platform efficiently tackles issues such as large dataset management, integration with other commonly used tools, data visualization, independently of the experimental workflow. This approach made the tool popular in many -omics using mass spectrometry irrespective of the size of identified molecules, that is, ranging from metabolites to large peptides, encompassing glycans and glycopeptides. Application to serum glycoprotein site occupancy using Skyline was reported early in (Hülsmeier et al. 2016). Since then, the use of the tool has spread both in glycomics and glycoproteomics.

6.3 Glycoproteomics Data Collection and Processing

6.3.1 Reference Databases

As hinted in the introduction, glycoproteomics is fast-growing and three appropriate databases have been released in recent years. In fact, the GlycoPOST repository (Watanabe et al. 2020) was designed for mass spectrometry data submission in relation to JPOST, the Japanese member of the ProteomeXchange Consortium. A database of curated glycoproteins and their associated glycans called GlyConnect (Alocci et al. 2019) includes many entries describing glycosylated

immunoglobulins. This database is reciprocally cross-linked with UniProt (The UniProt Consortium 2019), the reference protein sequence database, and feeds glycomics data to GlyGen (York et al. 2020), the recently released US portal for glycoscience.

6.3.2 Identification Software Tools

The topic of automatic intact glycopeptide identification is regularly reviewed (Thaysen-Andersen and Packer 2014; Hu et al. 2017; Cao et al. 2020; Abrahams et al. 2020), following the abundant production of new tools that warrant frequent updates of the catalog. This chapter is not destined to add to this set of reviews.

One of the key points in running glycopeptide identification software is the selection of the glycan composition file upon which the identification of the glycan moiety is based. In the majority of cases, the composition file can be qualified as "empirical" since its definition relies on glycan data collected from the literature or from databases/repositories. It is, however, an unevenly flexible definition. In some software, it is a modifiable parameter while in others it is not. Alternatively, the knowledge of glycan biosynthesis can be used to generate expected glycans. For example, MAGIC (Lih et al. 2016) starts with 29 monosaccharides to be combined according to enzymatic rules and in a similar way, 19 groups of seed structures are defined in GlycoPAT (Liu et al. 2017). In this way, the composition file is "theoretical" as the result of potential enzymatic activity. This definition is also flexible since both the enzymes or the monosaccharides can be removed or constrained.

Most classical proteomics search engines accommodate the selection of glyco-based modifications from a collection they provide. For example, in principle, Mascot (Savitski et al. 2011) relies on the UniMod subset of glycans (Creasy and Cottrell 2004). The user may also use in-house definitions as in (Bollineni et al. 2018), where a few hundred compositions were built from a customized and systematic addition of monosaccharides prior to using the Mascot engine. Note that instrument vendors also provide a glycopeptide identification component in the software suites that are usually available for proteomics data analysis such as ProteomeDiscover by Thermo or ProteinScape at Bruker, but the information on glycan composition files is not easily accessible.

Proteomics software developers, in particular when posttranslational modifications are accounted for, have already acknowledged that the performance of search engines correlates with limiting the search space (Schwämmle et al. 2015). Similar observations have been made in glycoproteomics and this viewpoint, also promoted here, is strongly advocated in (Khatri et al. 2017).

6.3.3 *Quantification Software Tools*

Most glycopeptide identification tools have some sort of built-in quantification module, either based on a spectral count or MS1 feature intensity or integrated area. While these modules are often sufficient to obtain a rough estimation of (relative) glycopeptide abundance, more sophisticated software tools have been reported for accurate glycopeptide quantification in a high-throughput manner. As described above, Skyline is well suitable for the quantification of glycopeptide data and has excellent visualization capability that allows for data curation and representation (Pino et al. 2020). LacyTools is an automated data processing tool for high-throughput analysis of LC-MS glycoproteomics data and is suitable for the simultaneous analysis and quality control of thousands of samples (Jansen et al. 2016). Both tools feature important options such as quality control based on isotopic pattern matching, separate integration of different charge states, and a flexible inclusion of isotope signals.

6.4 Data Integration with Other Omics

In the early years of bioinformatics in the 1980s when the term itself was hardly used, the main focus was on collecting and comparing gene and protein sequences. A handful of resources created then are still available now and this is the case of the international ImMunoGeneTics information system (IMGT), created in 1989 by Marie-Paule Lefranc in Montpellier, France. This resource remains a unique gene-centric reference in immunogenetics and immunoinformatics (Lefranc et al. 2015). IMGT/3Dstructure-DB is a subsection of the IMGT platform where Ig three-dimensional information is included. This data is cross-referencing the Protein Data Bank (PDB) (wwPDB consortium et al. 2019), where glycan information is symbolically represented in protein structures using a 3D version (Sehnal and Grant 2019) of the Symbol Nomenclature for Glycans (SNFG) depiction increasingly adopted in publications and textbooks (Varki et al. 2015).

Since the PDB has long considered glycans decorating glycoproteins as "ligands," IMGT lists "glucid" as potential ligands of immunoglobulins. IMGT/3Dstructure-DB (version 4.12.3) contains 21 entries where 3D structures include information on attached glycans. Figure 6.2 summarizes the steps of the search and illustrates the possible visualization of attached glycans with the example of the PDB entry 5 K65 of VEGF binding IgG1-Fc (Fcab CT6), as recorded from (Lobner et al. 2017).

The recently introduced feature of the PDB clearly showing the presence of a glycan on the 3D representation of the protein is the first step toward bringing together different data sources that cover complementary information on immunoglobulins. Further work in this direction is needed to reach a better level of integration between genomics and glycomics.

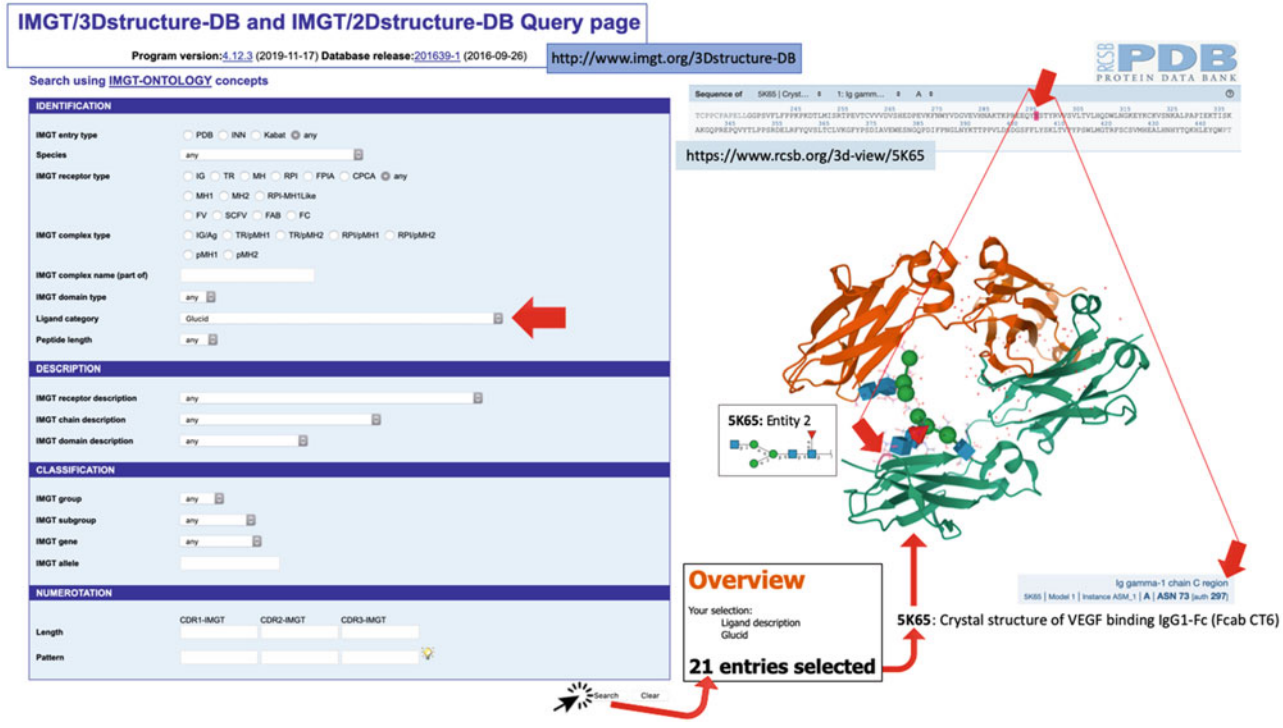


Fig. 6.2 Potential data integration with IMGT. The IMGT platform integrates genetic, structural, and functional information about immunoglobulins. It is split into several sections, one of which is devoted to 3D structures. Querying this section by ligand category with the "glucid" term (central red arrow) outputs a list of 21 entries with cross-links to the Protein Data Bank (PDB). The new interface of the PDB highlights the glycan in the 3D structure with the recommended SNFG notation

6.5 Practical Examples

This part describes the stepwise processing of glycomic and glycoproteomic Ig MS data, as well as the visualization of selected glycoproteomic identification results. Supplementary information is provided in the Notes section (notes 9–12), at the end of the chapter.

6.5.1 Glycomic Data Processing

The glycomic approach to Ig *N*-glycan characterization is traditionally performed after isolating the Ig of interest from its complex matrix by affinity chromatography. Subsequently, the *N*-glycans are released from their carrier protein using the enzyme Peptide-*N*-Glycosidase F (PNGase F). Prior to MS analysis, glycans may be subjected to permethylation, reduction, or reducing end labeling to increase sensitivity and specificity. Analysis of released glycans allows for in-depth structural elucidation and isomer separation, but it lacks information on the site of glycosylation. Therefore, for Ig-specific glycan analysis at the glycomics level, protein purity is of utmost importance. The glycomics data analysis protocol outlined below allows for the structural determination of reduced glycans released from human IgG. The workflow is illustrated in Fig. 6.3.

6.5.1.1 Software Required

1. *Manual inspection of raw data*: Bruker Daltonic Software (Bruker DALTONIK GmbH, Bremen, Germany) was used since the example data was generated with a Bruker instrument (amaZon speed Electron-transfer dissociation (ETD) ion trap mass spectrometer).
2. *Data format conversion*: MSConvertGUI (ProteoWizard 3.0 suite, http://proteowizard.sourceforge.net/)
3. *Determination of glycan composition from the experimentally determined precursor mass*: GlycoMod (https://web.expasy.org/glycomod/) that works well for both free and derivatized glycan as well as for glycopeptides.
4. *Interpretation of tandem MS spectra* (MS/MS spectra): GlycoWorkbench (https://code.google.com/archive/p/glycoworkbench/)
5. *Quantification*: Skyline (64-Bit) 20.2.0.343 (https://skyline.ms/project/home/begin.view?)

6.5.1.2 Glycan Composition Determination

1. Determine the experimental glycan masses. Convert any doubly and triply charged m/z values to singly charged or neutral mass. Bruker Data Analysis

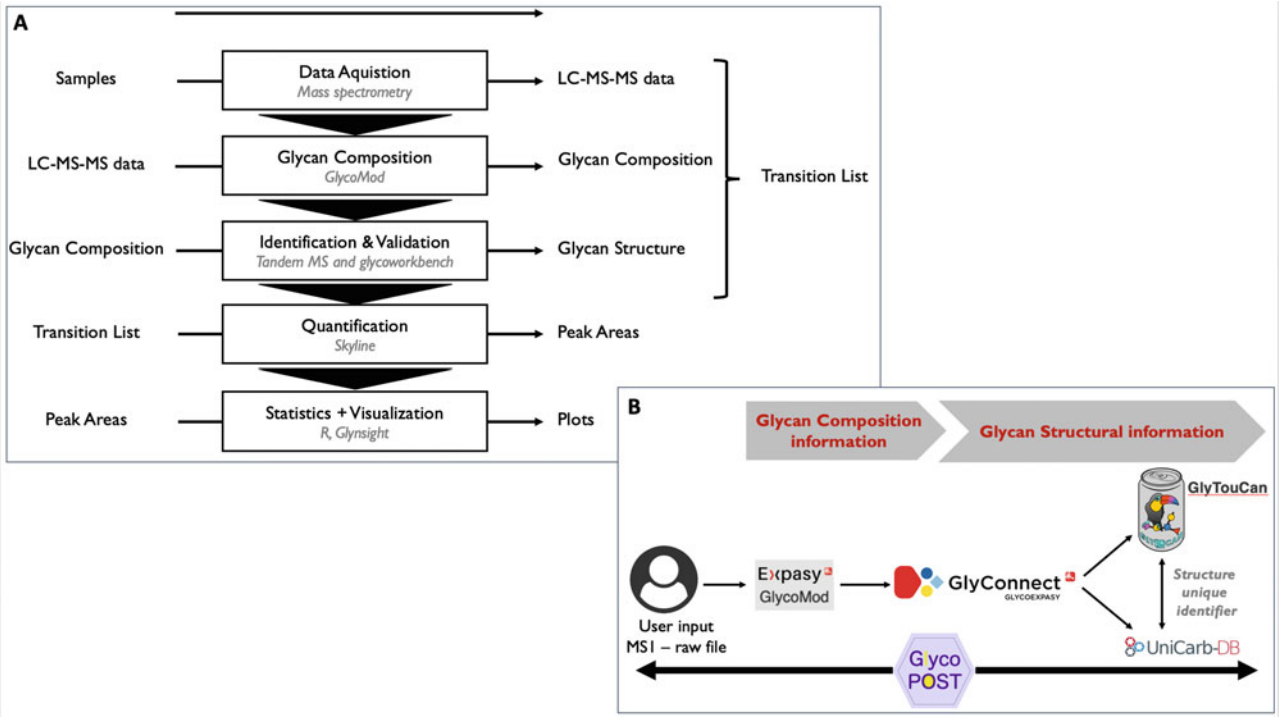


Fig. 6.3 Glycomics workflow. (**a**) The glycomics data analysis was divided into five major procedures. The raw data is acquired on the LC-MS instrument resulting in tandem MS spectra and retention time for each precursor mass selected for fragmentation. The MS1 masses allow for the determination of glycan composition followed by glycan structure identification and validation. The quantification can be performed using Skyline and peak areas can be exported and readily visualized using either R or various other visualization tools such as Glynsight available on the Expasy server. (**b**) MS raw files can be submitted to GlycoPOST including experimental metadata, glycan identification, and quantification results. Glycan composition information can be processed to obtain structural information

software allows for easy deconvolution of the precursor masses. For other instruments, a vendor-specific software package can be used for this purpose. As described below, KNIME (KNIME GmbH) with OpenMS 2.3 can be used for MS1 peak picking and deconvolution.

2. Open the web-based GlycoMod tool with the following parameters:
   (a) All mass values: Monoisotopic
   (b) Mass tolerance: ±0.2 Da (can be adjusted depending upon the mass accuracy of the instrument)
   (c) Ion mode and adduct: Neutral [M]
   (d) *N*-linked oligosaccharide: Reduced oligosaccharides
   (e) Monosaccharide residues present (if known): Underivatized and input the range if known
3. Enter the list of experimental masses.
4. At the bottom, click start GlycoMod.
5. A list of possible glycan compositions is calculated for each mass imputed with a link to GlyConnect.
6. Click on GlyConnect to view possible isomers where each structure is cross-referenced to GlyTouCan and UniCarb-DB.

6.5.1.3 Annotation of MS and MS/MS Glycomics Spectra

1. Launch GlycoWorkbench.
2. Draw the proposed glycan structure using the GlycanBuilder for each precursor mass. Theoretical m/z values can be calculated by configuring the parameters in the "Mass options" available under the "Edit" tab.
3. In silico fragmentation can be computed for selected structures using "compute fragments for selected structure" available in the "Tools" tab.
4. In the "Fragmentation options" pop-up window offers options to select specifications for fragmentation.
5. The results are displayed in a tabular form in the side panel containing all predicted fragments. Different fragments with the same exact mass are represented separately within the details tab. Whereas, the Summary tab contains a condensed form of the information presented within the detail tab.
6. Glycan structures can then be assigned based on glycan fragmentation pathways, in silico fragmentation, observed fragmentation, and precursor mass.
7. Additionally, experimental MS/MS peak lists can also be associated with the computed peak list.
8. Copy the list of fragments m/z and intensities directly from the MS software or export the list first to a spreadsheet using, for example, Microsoft Excel.
9. Next, go to Tools → Annotation → Annotate peaks with fragments from selected structures. A Fragment option window will appear. After setting the parameter, click OK. Then the associated peak list will appear in the right panel.

6.5.1.4 Convert Raw Mass Spectrometry Data to mzXML

1. Open MSconvertGUI.
2. Select the list of files that needs conversion and select the Output directory for the converted files.
3. In the options box below the output directory, adjust the settings to output format = "mzXML," Binary encoding precision = "64-bit," and check the boxes next to "write index," "use zlib compression," and "TPP compatibility."
4. In the filter box, select the subset option and do not change the settings that pop up (default setting).
5. At the bottom right corner, click "start," and wait for your files to finish converting to mzXML.

6.5.1.5 Skyline for Glycomics Quantitation

1. Open Skyline and select either the molecule interface or the mixed interface.
2. Go to the "Setting" tab and select "Transition settings."
3. In the pop-up window, select "Instrument" tab and input Min and Max m/z.
4. Next, in the "Full-Scan" tab, input/select the following parameters depending on the type of instrument used: (a) Isotope peak included (count), (b) Precursor mass analyzer (QIT), (c) Peaks (3), and (d) Resolution (0.3 m/z). The values in parentheses indicate the value used for the analysis.
5. At the bottom, select "Include all matching scans" and click OK.

6.5.1.6 Setting Up Transition List

1. Go to the "Edit" tab, and select Insert → "Transition list."
2. In the pop-up window, select "Molecules" at the bottom and click on the "Columns" button.
3. Select the required columns to set up the transition list. Be sure to include at least (a) Molecule List Name, (b) Precursor Name, (c) Precursor m/z, (d) precursor charge, (e) Explicit retention time, and (f) Explicit retention time window.
4. For large lists of glycans, it is advisable to create the "Transition list" in Excel.
5. The "Transition list" can be easily copied/imported to Skyline using File → Import → "Transition list." Once successful, you will see the Transitions in the Target menu.
6. Now save the File before importing the results (converted mzXML files).
7. Wait for the import to complete. Skyline integrates the area under the curve based on the information provided in the "Transition List." Check if the correct peaks are integrated for each glycan. If required, manual integration can be performed by clicking and dragging the integration window beneath the *x*-axis.

8. Skyline offers various parameters for the quality control of the integrated signals, including mass accuracy, isotopic pattern matching, and retention time which can be conveniently plotted for each sample.
9. Results can be exported using Files → Export → Report function as CSV file.

6.5.2 *Glycopeptide Data Processing for Enriched Immunoglobulins*

The analysis of Ig glycosylation at the glycopeptide level has the advantage that protein-, site-, Ig isotype- and often even subclass-specificity are maintained (Momčilović et al. 2020). General workflows include the affinity purification of the Igs of interest, their tryptic cleavage, and the analysis of the resulting (glyco)-peptide mixture by reverse-phase (RP) liquid chromatography (LC) coupled to high-resolution mass spectrometry (MS). The RP-LC separation of (Ig) glycopeptides results in the retention time clustering of analytes with the same peptide portion but different glycan moieties. Often, distinct retention time clusters are also obtained for glycopeptides with varying degrees of sialylation. Based on this very characteristic and predictable elution behavior of glycopeptides, software packages were developed for the identification [GlycopeptideGraphMS (Choo et al. 2019)] and targeted quantification and quality control [LaCyTools (Jansen et al. 2016)] of glycopeptides derived from purified proteins. Combining these tools with MS2-aided glycopeptide validation resulted in a workflow for the semiautomated identification and quantification of glycopeptides that is highly suitable for high-throughput Ig glycosylation analysis (Lippold et al. 2020). The protocol outlined below is adjusted for data [publicly available via (Bern et al. 2012)] obtained by the RP LC-Orbitrap-MS/MS analysis of tryptic IgG and IgA glycopeptides, Fig. 6.4 (Glycoproteomics workflow).

6.5.2.1 Software Required

1. *Manual inspection of the raw data*: instrumentation proprietary software. In this case, Xcalibur (v. 2.2, Thermo Fisher Scientific).
2. *MS2-based glycopeptide identification*: software package for automated MS/MS assignment of glycopeptides [example can be found in (Abrahams et al. 2020)]. In this case, PMI-Byonic (v. 3.7.13, Protein Metrics), (Bern et al. 2012) Fig. 6.4a.
3. *Data format conversion*: MSConvertGUI (ProteoWizard 3.0 suite, http://proteowizard.sourceforge.net/) (Kessner et al. 2008), Fig. 6.4b.
4. *MS1 peak picking and deconvolution*: KNIME (KNIME GmbH) with OpenMS 2.3 (https://www.openms.de) (Pfeuffer et al. 2017). Details on the installation of these, along with a dedicated workflow to prepare GlycopeptideGraphMS inputs are available as part of the GlycopeptideGraphMS download. (workflow: KNIME_OPENMS_GraphMS_Preprocessing_120318, Fig. 6.4c; https://bitbucket.org/glycoaddict/glycopeptidegraphms/src/master/).

5. *MS1-based glycopeptide identification*: GlycopeptideGraphMS, Fig. 6.4c (v. 2.06, https://bitbucket.org/glycoaddict/glycopeptidegraphms/src/master/).
6. *Glycopeptide quantification and quality control* (QC): LaCyTools, Fig. 6.4d (v 2.0.1, https://git.lumc.nl/cpm/lacytools).
7. Python 3 with the Anaconda package (https://www.anaconda.com/).

6.5.2.2 Glycosylation Site Identification Based on MS2

Byonic can be used to identify the peptides that are present in samples as well as validating the presence of glycopeptides (Fig. 6.4a). There are three inputs: raw files, a protein database, and a glycan database (in this case, Byonic proprietary *N*-linked database).

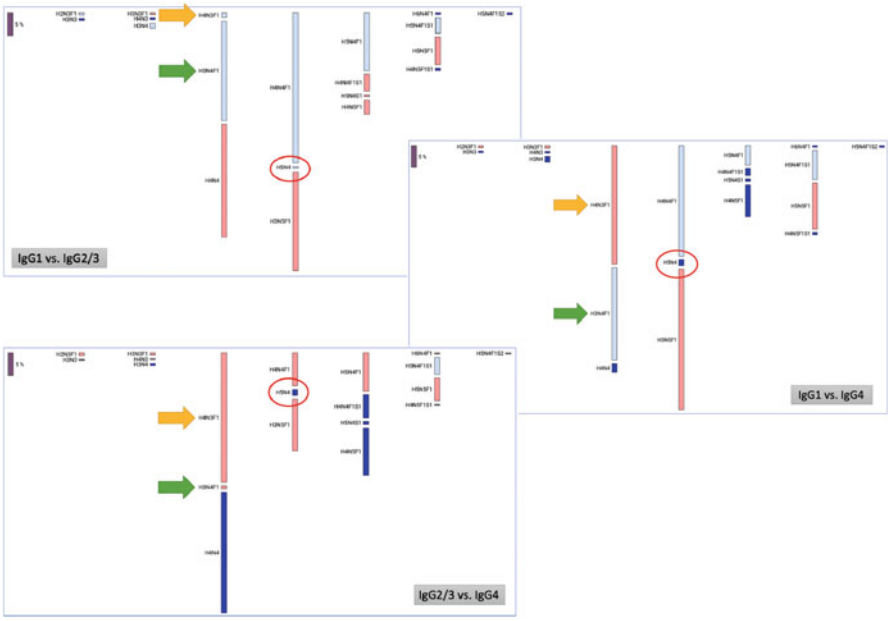


Fig. 6.4 Glycoproteomics workflow. The glycoproteomics dataset was subjected to (**a**) glycosylation site identification by MS2 and (**b**) MS1 feature selection and deconvolution. (**c**) The outcomes of these processes were combined using GlycopeptideGrapMS for extensive glycoform identification and visualization. (**d**) The identified glycopeptides were subjected to a QC step and quantified using LaCyTools

1. Run PMI-Byonic on the raw LC-MS/MS data files with the Homo sapiens UniProt database including 71,591 protein sequences (20,205 from Swiss-Prot and 51,386 from TrEMBL at time of writing). Byonic has an inbuilt glycan database for putative glycan assignments. In this example the "N-glycan 309 mammalian no sodium" glycan list is used. (see Note 1). Individual options are set to:

   (a) C-terminal cleavage of lysine and arginine: true
   (b) Maximum number of missed cleavages: 2
   (c) Precursor mass tolerance: 10 ppm
   (d) Fragment ion mass tolerance: 20 ppm
   (e) Fixed modification: cysteine carbamidomethylation
   (f) Variable modification: methionine oxidation

2. Output from this step is a list of polypeptide cleavage products and glycopeptide identifications. Identify those products of the protein(s) of interest. Include glycopeptide identifications with a Byonic score above 200 (see Note 2) (Fig. 6.4a).

6.5.2.3 Glycoform Identification Based on MS1

GlycopeptideGraphMS can be further used in the identification process to cluster the identified glycopeptides by retention time. These nodes can reveal information on the underlying proteins and glycosylation sites. Advanced (and/or customized) options can be investigated using GlycopeptideGraphMS manual (https://bitbucket.org/glycoaddict/glycopeptidegraphms/).

1. A preprocessing step is required to create input for GlycopeptideGraphMS. Convert the raw data to the mzML format using MSConvertGUI with the following settings:

   (a) Binary encoding precision: 64-bit
   (b) Filters: subset – MS level: 1-1
   (c) Check: write index, use zlib compression, TPP compatibility

2. Load the obtained mzML file in the KNIME OpenMS workflow and define an output directory. Adjust the default settings in the feature finder to account for the m/z range (400–3500) and expected charge states (2–7). Run the file filter, peak picking, feature finder, and de-charging node (Fig. 6.4b). Adjust these settings according to instrumentation.
3. Load the obtained .csv file from the KNIME OpenMS workflow into GlycopeptideGraphMS and format it into the consensus format needed for further processing using the "Format consensusXML" function.
4. Perform the GlycopeptideGraphMS analysis with the following settings (only values deviating from the default are mentioned):

(a) Mass deviation tolerance: 0.02 Da
(b) Maximum subgroup degree: 1 (the lower the value, the higher the noise)
(c) Composition search blocks (see Note 3):

(i) Hexose (Hex, 162.0528 Da, max. 30 s retention time (RT) difference)
(ii) *N*-Acetylhexosamine (HexNAc, 203.0794 Da, max. 30 s RT difference)
(iii) Hexose + *N*-acetylhexosamine (HexHexNAc, 365.1322 Da, max. 30 s RT difference)
(iv) Deoxyhexose (Fuc, 146.0579 Da, max. 20 s RT difference)
(v) *N*-Acetylneuraminic acid (NeuAc, 291.0954 Da, max. 120 s RT difference)

5. The obtained output.csv provides a list of all nodes identified in so-called "subgraphs" (glycopeptides that share the same peptide portion and differ in their glycan content). Entries (rows) with the same value under "refnode" belong to the same subgraph. Use the m/z (mz_node) and retention time (rt_node) of each node to match them to glycopeptides identified during the MS2-based identification (Sect. 5.2.2).
6. Open the "reference node XLSX" which lists one node per identified subgraph. Populate each entry by one MS2-confirmed node per subgraph (see Note 4 and Note 5), indicating the node number (node), m/z (mz_node), retention time (rt_node), the number of the subgraph node with the lowest mass with its m/z (refnode; mz_refnode), the number of the respective monosaccharides present in the identified glycopeptide and the peptide sequence of the identified glycopeptide, as illustrated in Table 6.2.
7. Use the graphical representation option in GlycopeptideGraphMS to visualize the identified subgraphs. Closely related clusters help to identify (unexpected) peptide modifications and glycoforms, Fig. 6.4c which can be used to re-run the glycopeptide (Byonic) search and optimize the output. It also allows the identification of unexpected peptide modifications, such as partial cysteine oxidation (identified on the IgA data in this example).
8. Run the composition predictor in GlycopeptideGraphMS (input: reference node XLSX and output.csv) to obtain a full list of glycoforms, based on the MS1 LC-MS features.
9. Remove the glycoforms from the predicted composition output of GlycopeptideGraphMS that contain negative values in the final glycoform composition for one or more of the composition search blocks (illogical nodes). Use the graphical representation to remove the glycoforms with logical compositions that were only connected to the subgraph via an illogical node.

Table 6.2 Description of an MS2-confirmed node per subgraph, following the GlycopeptideGraphMS analysis

Subgraph	node	mz_node	rt_node	Refnode	mz_refnode	N	H	F	S	Pep
6	989	2658.155	1769.709	989	2658.155	3	4	1	1	SLHVPGLNK

6.5.2.4 Targeted Glycopeptide Quantification

1. Convert the raw data (see Note 6) to the mzXML format using MSConvertGUI with the following settings:
   (a) Binary encoding precision: 64-bit
   (b) Filters: subset – MS level: 1-1
   (c) Check: write index, use zlib compression, TPP compatibility
2. Place the LaCyTools master folder in the working directory.
3. Based on the peptide fragments identified in the previous steps, create text files for the peptide "building blocks" representing the peptide potions, following the format below (example for the IgA2 glycopeptide covering site Asn337). Save the text file as [peptidename].block file and place it in the blocks folder in the LacyTools master folder (Fig. 6.4d).
   TPL.block format:
   (a) mass
   957.5495
   (b) available_for_charge_carrier 0
   (c) carbons 42
   (d) hydrogens 75
   (e) nitrogens 11
   (f) oxygens 14
   (g) sulfurs 0
4. When there is more than one data file to quantify, pick one as the reference file. Create a tab-delimited text file (LaCyTools alignment file) listing the m/z of a minimum of five highly confident (see Note 7) glycopeptide compositions covering the entire retention time range in the first column and the retention time of these features in the reference file in the second column.
5. Create a tab-delimited text file (LaCyTools analyte list) containing the following:
   (a) Column 1: all identified glycopeptides, using the annotation TPL1H5N4F1S1 for the peptide covering IgA2 site Asn337 (TPL), carrying a glycan with five hexoses (H), four *N*-acetylhexosamines (N), one fucose (F), and one sialic acid (S)
   (b) Column 2: the retention time of the center of the LaCyTools glycopeptide cluster the analyte belongs to
   (c) Column 3: the integration m/z window
   (d) Column 4: the glycopeptide cluster retention time window
   (e) Column 5: the minimum charge state for the analyte
   (f) Column 6: the maximum charge state for the analyte
   A dedicated python script is available from [4] to create the LaCyTools analyte list based on the GlycopeptideGraphMS output.

6. In LaCyTools, open the Settings tab and adjust as follows:
   (a) Alignment time window and Alignment mass window: match the variation in retention time over the different runs and the mass accuracy of the analysis, respectively
   (b) Sum spectrum resolution: the average number of data points per 1 m/z in the MS1 spectra
   (c) Charge carrier: proton
   (d) Extraction window padding: 0
   (e) Minimum isotopic fraction: 0.75
   (f) Background detection window: 10 (see Note 8).
7. Open the Batch process tab, select the created LaCyTools alignment file under Alignment file and the LaCyTools analyte list under Reference file. Choose the directory where the .mzXML files are stored and select the following output format: Analyte Intensity, per charge state, background subtracted; Alignment QC; Analyte QC. Run the batch process.
8. The obtained Summary file contains the absolute intensities and QC parameters (mass accuracy, isotopic pattern QC; IPQ, and signal-to-noise; S/N) of all targeted glycopeptides in the selected charge states. In Excel: remove charge states of analytes with an S/N < 9, an IPQ > 0.25 and an absolute ppm error > 10. Sum the areas of the remaining charge states per analyte and correct the total for the exact fraction of the isotope pattern integrated for the respective analyte.
9. To obtain the final data, perform total area normalization per glycopeptide cluster (subgraph). For each data file, sum the absolute intensities of all glycoforms per peptide portion and divide each individual glycopeptide of this cluster by the sum. The obtained values can be used for data visualization and statistical analysis (Fig. 6.4d).

Using this protocol glycopeptide data was quantified for IgG1, IgG2/3, IgG4, IgA2 Asn337 and the joining chain Asn49 (Lippold et al. 2020).

6.5.3 Visualizing Profiles

We rely here on a so-called condensed notation: H for hexose, N for hexosamine, F for fucose, and S for sialic acid. Additionally, small letters designate modifications such as “a” for acetylation, “p” for phosphorylation, and “s” for sulfation, for the most frequent.

6.5.3.1 Structural Dependencies Brought Out by GlyConnect Compozitor

The output of most search engines identifying glycans or intact glycopeptides is usually provided as a list or in a table where the items or the rows are independent. However, the glycan structures or the compositions associated with peptides share

common substructures. The GlyConnect Compozitor tool was developed to visualize these dependencies (Robin et al. 2020). Moreover, this web interface provides the option of comparing experimental results with the content of the GlyConnect database.

The glycoproteomics results of Sect. 5.2 for Ig gamma 4 were selected. Nine compositions were identified on peptide EEQFNSTYR (Asn177 in the corresponding UniProt P01861 entry):

H2N3F1
H3N3F1
H4N3F1
H3N4F1
H4N4F1
H3N5F1
H5N4F1
H5N5F1
H5N4F1S1

To bring out dependencies, Compozitor maps compositions related by shared monosaccharide counts into a graph that eases glycome visualization and comparison. When available, each composition is associated with defined structures. Here is how to generate graphs:

1. Go to glyconnect.expasy.org.
2. On the homepage, click on the orange PROTEIN button to open the Protein View of the database.
3. In the Protein View page, type "P01861" in the search window.
4. In the output of two entries, select/click on entry 770 (Id column), i.e., human immunoglobulin heavy constant gamma 4, for display.
5. The protein page shows all glycan structures reported to be *N*-linked to Asn-177 in ten different references. On the right side of the page, click on the Compozitor link.
6. The Compozitor search fields are pre-filled with the details of the 770 entry. In this case, the protein name is "immunoglobulin heavy constant gamma 4" and the Species is "Homo sapiens," shown in the "Protein" tab.
7. Click on the "Add to selection" button and observe that 33 compositions are recorded in the GlyConnect database, ready for display as "Selection A." Then, move to the "Custom" tab by clicking on it.
8. Copy-paste the nine compositions listed above in the input window. The glycan type is *N*-linked by default. Type "experimental" in the "selection label" window in order to name your input set of compositions.
9. Click again on the "Add to selection" button and observe that the nine compositions are now saved as "Selection B."
10. Display the graph of connected compositions by clicking on the "Compute graph" button.

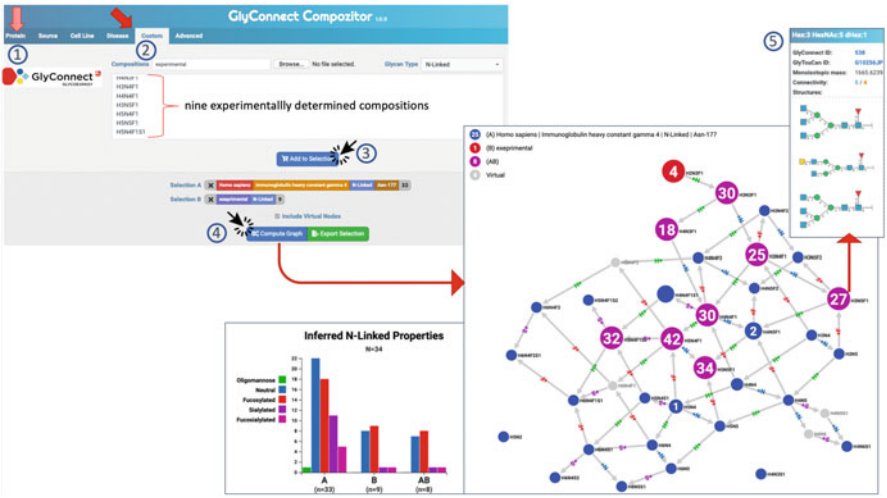


Fig. 6.5 Compozitor output of IgG4 data. The GlyConnect Compozitor tool is accessed directly from the database when querying it to visualize the *N*-glycome of human IgG4. This step is numbered 1 and the data is visualized in the Protein tab. It is labeled "Selection A." Step #2 consists of selecting the Custom tab to input the nine experimentally identified compositions in the glycoproteomics screening of IgG4. During step #3 the input data is recorded as "Selection B" and in step #4, the Compozitor graph is output. Magenta nodes correspond to the match between experimental data and the database content. A window with links to potential structures pops up by mousing over a node in step #5

The graph appears below and shows a well-connected set of 33 + 9 composition nodes that are distinguished by color: blue for the 33 compositions recorded in the database (Selection A), red for the nine input experimental set (Selection B), and magenta for the overlap between the two, as shown in Fig. 6.5. A number contained in a node represents how many glycan structure entries match this composition in GlyConnect. For example, H3N5F1 matched 27 partially or fully defined structures. This information is accessible by mousing over the node to prompt a caption showing the potential structures as shown in Fig. 6.5. The user can then explore the possibilities by clicking on the structure thereby opening the corresponding GlyConnect glycan structure page. The graph shows that the experimental compositions match exactly the mostly documented structures in the database (nodes with highest numbers).

Compozitor allows for "virtual nodes" (in gray) when only one intermediary step is needed to connect isolated nodes. If not for these, the graph would be more scattered.

6.5.3.2 Comparing Profiles with Glynsight

Again, the results of glycopeptide quantification generated by the workflow of Sect. 5.2 were recorded and input in the Glynsight tool (Alocci et al. 2018) for comparing

the expression profiles of each of the IgGs. An example of file in the correct .csv (comma separated values) input format is shown below for IgG4:

```
composition,quantification,
H2N3F1,0.55,
H3N3F1,1.74,
H3N4F1,0.59,
H3N5F1,37.19,
H4N3F1,28.24,
H4N4F1,6.99,
H5N4F1,12.01,
H5N4F1S1,1.04,
H5N5F1,11.64,
```

1. Go to https://glycoproteome.expasy.org/glynsight;
2. In the left menu, click on "Manage experiments" and click on "Import experiments" in the list of tasks that is displayed. Use the "upload files" button to import the .csv files such as the one shown above. The easiest is to give a meaningful name to the file, such as IgG4.csv for the example above.
3. The name of each of the uploaded files is shown in the left menu under "Experiment list." Click on each name of experiment to see the corresponding profile displayed as a bar chart, each bar representing a composition. These are incrementally ordered and structured in columns. All compositions featuring in a column have a constant overall number of monosaccharides. The first column of the profile here contains H3N3 and H2N3F1 (six monosaccharides in total), the next column contains compositions with an overall count of seven monosaccharides, etc. Changing from one experiment to the next immediately brings out the variations.
4. The default display mode is "individual," showing one experiment at a time. To compare profiles, select "differential display" in the Display mode section of the left menu. In this mode, the list of experiments in the left side menu is duplicated to provide the user with the option of selecting an experiment in each of the two lists.
5. Compare IgG1 with IgG2/3, then with IgG4, etc.

Figure 6.6 shows the profile comparisons. Glynsight shows red and blue bars whose height for each composition represents the result of the subtraction. For any given composition, if its intensity in the first experiment selected is higher than in the second, then the bar is blue. In the opposite situation, the bar is red. Brighter blue (resp. red) highlights differences arising from compositions unique to the first (resp. second) experiment while toned down blue (resp. red) shows differences arising from compositions expressed in both experiments but higher in the first (resp. second) than in the second. Grey corresponds to glycans present in other experiments but not in those currently displayed.

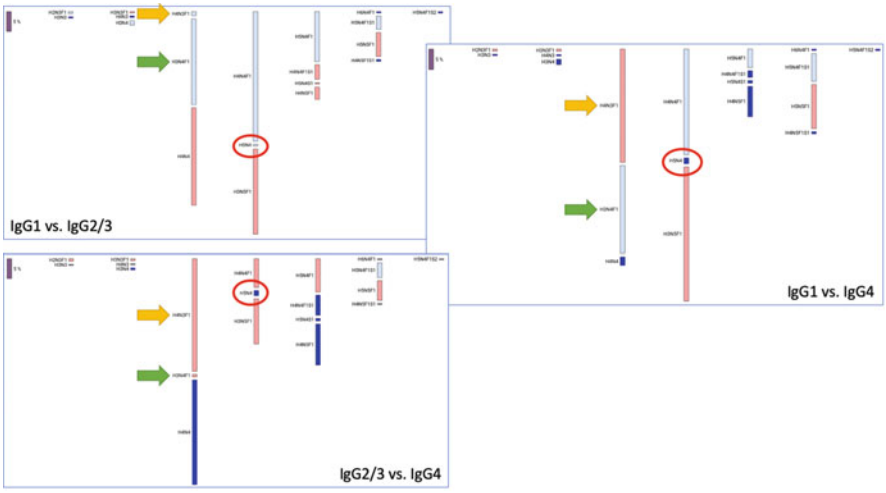


Fig. 6.6 Glynsight comparison of profiles. The comparative profiles of IgG1, IgG2/3, and IgG4 with respect to one another as displayed by Glynsight. Blue (resp. red) bars indicate that expression is higher (resp. lower) in the first profile than in the second. Brighter blue (resp. red) shows differences arising from compositions unique to the first (resp. second) profile while toned down blue (resp. red) shows differences arising from compositions expressed in both profiles but higher in the first (resp. second) than in the second. Green arrows single out the expression of H3N4F1 underexpressed in IgG1, but roughly the same in IgG2/3 and IgG4. Yellow arrows single out the expression of H4N3F1 overexpressed in IgG4 but roughly the same in IgG1 and IgG2/3. Red circles show the constant expression of H5N4 across all sets

The large bars reflect significant differences between the two profiles. The screen capture of the figure is not as visual as when using the software live, since the dynamics of change are not transcribed. Nonetheless, variations are observable. For example, H3N4F1 is overexpressed in IgG1 with respect to both IgG2/3 and IgG4, while it is the same between IgG2/3 and IgG4 (see green arrows in Fig. 6.6). In contrast, H4N3F1 is over overexpressed in IgG4 with respect to both IgG1 and IgG2/3, while it is the same IgG1 and IgG2/3 (see yellow arrows). Another example of constant expression across all sets is that of H5N4 (red circles).

6.6 Conclusion

This chapter has demonstrated the extent of available bioinformatics resources for studying the glycosylation of immunoglobins. On the one hand, both generic and specific databases are interconnected to reflect the growing body of literature on Ig glycosylation and its functional influence. The therapeutic importance of these proteins warrants a special section soon to be developed in the GlyConnect database. On the other hand, the cited software tools have enabled high-throughput and robust analysis, especially for IgG and IgA. In fact, Ig(G)s have quite a head start, compared

to other glycoproteins and many publications already reveal IgG *N*-glycome variations as a reflection of disease states (see recent (Clerc et al. 2018; Martin et al. 2020) for examples). Although the same basic principles apply for the analysis of other glycoproteins, more developments and streamlining of resources are warranted, especially when shifting from isolated proteins to complex samples.

Notes

Glycopeptide data analysis

1. This *N*-glycan library is a good start for Ig samples but should be used with caution as it is far from complete regarding reported *N*-glycan structures. The glycan library used in Byonic can be fine-tuned in an iterative manner after Step 7 in "Glycoform identification based on MS1." In addition, GlyConnect (https://glyconnect.expasy.org/) can be used to construct a protein-specific library based on literature knowledge.
2. According to the Byonic user manual (https://www.proteinmetrics.com/support-information/), the Byonic score ranges from 0 to 1000. A score of 300 is considered good, 400 is very good and scores over 500 are almost sure to be correct. In our experience, lower scores may reflect confident glycopeptide identification, and as further QC will be performed both during the MS1-based identification and the quantification step, using a relatively low threshold score of 200 is justified here.
3. The max. RT differences are highly dependent on the chromatographic setup used and should be optimized based on the specific data analyzed. In general, when using C18 chromatography, the RT difference between two neutral monosaccharides is smaller than between a neutral and a charged monosaccharide.
4. In the likely scenario that more than one node per cluster could be assigned to a specific glycopeptide by MS2, choose the one with the highest confidence. Confidence can be based on Byonic score, manual evaluation of the MS2 data, and prior (literature) knowledge on the expected products.
5. In the scenario that no MS2 confirmed glycopeptide was obtained for any of the nodes in one subgraph, leave the row empty. This can indicate multiple things, including (1) no MS2 spectra were obtained for this glycopeptide cluster, in this case, a targeted MS2 run can be considered; (2) an unknown glycopeptide modification occurred, when this modification is identified Byonic can be run again including it in the search; (3) these nodes were erroneously identified as glycopeptide cluster, exclude the subgraph.
6. To obtain superior quantification precision, data integration should be performed on MS1-only data featuring a high number of data points per chromatographic feature. [4] However, data files including MS2 scans are also suitable, provided that there is a minimum of five MS1 data points per chromatographic feature.
7. While subgraphs or glycopeptide clusters are defined as a group of glycopeptides that share the same peptide portion and differ in their glycan content, in the current context separate LaCyTools glycopeptide clusters were defined for

glycopeptides with the same peptide portion but different sialic acid contents. In general, the more clusters defined, the longer LaCyTools processing time, but the lower the chance of analyte overlaps that interfere with the quantification.

8. Please find an in-depth explanation of all settings in [3] and [5]. The definition of the isotopic pattern QC (IPQ) changed between the initially published version of LaCyTools and the current and is now defined as the sum of the relative intensity deviation per isotope.

Visualization tools

9. Several of the images showing a Compozitor graph in figures have been slightly modified from the raw output to disentangle the network which is generated by the D3.js (https://d3js.org) library. This transformation is made easy since any node can be dragged wherever a user wishes in the space of the browser window by maintaining the mouse over it. Consequently, paths can be shortened or stretched upon user actions.
10. Compositions can be exported in a text file as soon as they have been added to the selection and before computing the graph ("Export selection" button). They can be selectively exported once the graph is computed ("Export" button). The graph can be exported in the .svg format.
11. The "Zoom in" and "Zoom out" buttons have obvious purposes and the former operation is supplemented by the "Zoom on" window located in the top right corner; typing a specific composition in this window will trigger zooming in and centering the graph on the corresponding node.
12. Mousing over a bar of the bar plot of glycan properties highlights all nodes that are counted in that bar to show where they are located in the graph.

Compliance with Ethical Standards

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Ethical Approval This chapter does not contain any studies with human participants performed by any of the authors.

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Kathirvel Alagesan declares that he has no conflict of interest.
Catherine Hayes declares that she has no conflict of interest.
Steffen Lippold declares that he has no conflict of interest.
Noortje de Haan declares that she has no conflict of interest.

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